# Introduction to CLI (10.23.2026)

## Cisco IOS CLI & Privileged EXEC Security — Class Notes

### 1. Cisco IOS Command Mode Flow

```
Router>                 User EXEC mode
  | enable
  ▼
Router#                 Privileged EXEC mode
  | configure terminal
  ▼
Router(config)#         Global configuration mode
```

- `Router>` — user EXEC mode
- `Router#` — privileged EXEC mode
- `Router(config)#` — global configuration mode
- `enable password` and `enable secret` are **global configuration commands**, so they must be entered at `Router(config)#`, not directly at `Router#`.

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### 2. Enable Password vs. Enable Secret

Both commands set the password requested when a user runs `enable` to enter privileged EXEC mode.

But they differ in security.

| Feature | `enable password` | `enable secret` |
|---|---|---|
| Prompted at `enable`? | Yes | Yes |
| Default storage | Clear text | Hashed |
| With `service password-encryption` | Type 7, weak/reversible | Unaffected |
| Security level | Weak | Strong |
| If both configured | Ignored | Takes precedence |
| Recommended? | Avoid | Use this |

#### Key point

If both are configured, **only the `enable secret` password is accepted** when entering privileged EXEC mode.

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### 3. Corrected Command Reference

| Command | Correct mode | Purpose |
|---|---|---|
| `enable` | User EXEC `>` | Enter privileged EXEC mode; prompts for enable secret/password |
| `configure terminal` | Privileged EXEC `#` | Enter global configuration mode |
| `enable password PASSWORD` | Global config `(config)#` | Set weak enable password |
| `enable secret PASSWORD` | Global config `(config)#` | Set strong hashed enable secret |
| `service password-encryption` | Global config `(config)#` | Encrypts clear-text passwords using weak type 7; does **not** protect `enable secret` |
| `no <command>` | Global config `(config)#` | Removes a configured command |
| `show running-config` | Privileged EXEC `#` | Display active configuration |
| `show startup-config` | Privileged EXEC `#` | Display saved configuration |
| `write memory` | Privileged EXEC `#` | Save configuration to NVRAM |
| `copy running-config startup-config` | Privileged EXEC `#` | Save configuration; preferred modern command |

#### Important correction to your notes

Your note had:

```
Router(config)# run privileged-exec-level-command
```

That is not a real Cisco IOS command.

From global configuration mode, you can run privileged EXEC commands using `do`:

```
Router(config)# do show running-config
Router(config)# do write memory
```

So replace that line with:

```
Router(config)# do show running-config
## runs a privileged EXEC command from global configuration mode
```

Also, `show`, `write`, and `copy` are privileged EXEC commands. From `(config)#`, use `do ...` or exit back to `#`.

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## 4. Console Connection Cable

**RJ45 console port → rollover cable**

- Cisco RJ45 console port uses a **rollover cable**.
- Often light blue, RJ45-to-RJ45, with a DB9/USB adapter.
- Not a crossover cable.
- Modern devices may also offer mini-USB/USB console ports.

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## 5. Quiz Answer Key

1. What kind of cable is used to connect to a Cisco device via the RJ45 console port?

   **a) Rollover cable**

2. You type `enable`, but the password is not accepted. What could be the problem?

   **c) Caps Lock is on**

   Real-world also check: wrong password, `enable secret` precedence, keyboard layout.

3. Most secure method to protect privileged EXEC mode?

   **a)** **`enable secret`**

4. If both `enable password` and `enable secret` are configured, what happens?

   **c) You must enter the `enable secret` only.**

5. Full version of `conf t`?

   **b)** **`configure terminal`**

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## 6. Extra Security Notes

- `service password-encryption` is **not strong security**. It uses reversible type 7 encryption.
- `enable secret` is automatically hashed when configured.
- Older IOS uses type 5 MD5; newer IOS supports stronger type 8/9 hashes. Where available, use stronger algorithms.
- You cannot recover an `enable secret` from the config; you can only reset/replace it.
- Prefer `enable secret` and avoid `enable password` entirely.

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## 7. Quick Memory Summary

```
enable password  = weak, clear text / type 7
enable secret    = strong, hashed, wins if both exist
enable prompt    = happens when you type "enable"
both commands    = entered in global config mode
show/save/copy   = privileged EXEC; from config use "do ..."
console cable    = rollover cable
```